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Automated machine learning achieves accurate water quality prediction with reduced parameter requirements

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Accurate water quality assessment is critical for environmental monitoring and public health. Conventional Water Quality Index (WQI) computation methods, however, often rely on numerous parameters and labor-intensive processes, thus limiting their practicality for rapid assessments. While Machine Learning (ML) offers promising alternatives, the development of high-performing models typically demands extensive expertise and computational resources. This study addresses the latter gap by leveraging Automated Machine Learning (AutoML), specifically the AutoGluon platform, to predict WQI from a reduced set of readily available water quality parameters. Our objectives were to (i) evaluate the predictive performance of AutoML with reduced inputs, (ii) assess model interpretability via feature importance, and (iii) propose an automated framework for efficient water quality monitoring. This study analyzed a 36-year dataset from Taiwan's national river water quality monitoring network, focusing on four parameters: electrical conductivity (EC), suspended solids (SS), water temperature (WT), and pH. A fully automated pipeline handled model selection, hyperparameter tuning, and ensemble construction by systematically testing multiple algorithm families and stacking strategies to determine the optimal setup for each parameter. This eliminated manual intervention and delivered reproducible, data-driven results that matched the distinct spatiotemporal patterns in the long-term records. Among the evaluated algorithms, ensemble-based tree models (CatBoost, Random Forest, and XGBoost) demonstrated superior performance, achieving a mean R^2 of 0.76 and low predictive errors. Feature importance analysis identified EC as the most influential predictor. Such results highlight the feasibility of reducing the number of input parameters without compromising prediction accuracy, thereby enabling faster and cost-effective water quality assessments.

The escalating pressures of urbanization, industrialization, and intensive agricultural practices have precipitated a significant deterioration of water quality worldwide, posing critical challenges to environmental sustainability and public health[1–4]. Coastal urban centers are particularly vulnerable, facing complex pollution dynamics driven by the influx of industrial effluents, domestic sewage, and agricultural runoff[5,6]. These dynamics necessitate robust, timely, and scalable monitoring frameworks that can inform effective management and policy interventions.

The magnitude of this crisis manifests differently across global contexts, affecting both developed and developing nations. Historically severe water pollution in Asia, such as the contamination of 70% of China's rivers and lakes in the early 2010 s, has driven significant remediation efforts, achieving 90.4% excellent-to-good water quality by 2024[7,8]. Despite such progress, substantial challenges persist: approximately 91 million people in India still lack access to clean water sources[9], while across Africa, Asia, and Latin America, water pollution has worsened significantly since the 1990s. Pathogen pollution affects approximately one-third of all rivers in

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these regions, placing over 300 million people at risk of life-threatening diseases such as cholera and typhoid[10]. In developing countries, inadequate infrastructure, limited funding, and insufficient technical capacity further compound monitoring challenges[11], thus underscoring the urgent need for accessible, cost-effective, and scalable assessment methodologies.

The complexity of contemporary water quality assessment extends beyond the scale of contamination to encompass the proliferation of multiple contaminant classes. In addition to conventional pollutants, emerging contaminants such as pharmaceuticals, personal care products, microplastics, and endocrine-disrupting compounds have attracted significant attention due to their potential ecological and health impacts[12,13]. These emerging pollutants often occur at trace concentrations yet can exert significant biological effects, necessitating advanced monitoring frameworks[12,14]. Such complexity is further compounded by spatiotemporal variability in contamination patterns, influenced by seasonal variations, hydrological dynamics, land-use changes, and anthropogenic activities[15,16]. Understanding the latter spatiotemporal dynamics is critical for developing adaptive monitoring strategies and targeted remediation efforts.

The above-described multifaceted pollution patterns translate into significant health implications that extend beyond acute exposure risks. Indeed, recent studies have developed comprehensive health risk assessment frameworks that quantify both carcinogenic and non-carcinogenic risks associated with waterborne contaminants[12,17]. Such assessments integrate exposure modeling, toxicological data, and population vulnerability factors to estimate disease burdens and inform water quality standards[18]. Central to developing effective interventions is understanding contamination sources through hydrogeochemical characterization and multivariate source apportionment techniques[18,19]. The latter approaches combine water-chemistry analysis with statistical modeling to distinguish natural geogenic processes from anthropogenic pollution sources, thereby enabling more targeted pollution-control measures.

Building on these diagnostic capabilities, addressing water quality challenges requires implementing effective remediation strategies. Recent advances in pollutant removal technologies have expanded the available toolkit, ranging from conventional physical-chemical processes to innovative approaches such as advanced oxidation, membrane filtration, adsorption on novel materials, and bioremediation[12,20]. The selection and optimization of appropriate treatment technologies depend on contaminant characteristics, water matrix properties, and site-specific constraints. To assess such complex decisions, Multi-Criteria Decision-Making (MCDM) models have emerged as valuable management tools, thereby enabling systematic evaluation of alternative strategies that consider technical, economic, environmental, and social criteria simultaneously. MCDM frameworks such as Analytic Hierarchy Process (AHP), TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution), and PROMETHEE (Preference Ranking Organization Method for Enrichment Evaluations) have been successfully applied to prioritize monitoring stations, select treatment technologies, and allocate water resources, proving particularly valuable in resource-constrained settings where trade-offs among competing objectives must be explicitly considered[21].

The Water Quality Index (WQI) is widely used globally as an integrative metric that combines multiple physicochemical and biological parameters to provide an overall assessment of water quality[22,23]. It facilitates temporal and spatial trend analyses, pollution source identification, and the evaluation of the effectiveness of regulatory measures, thus serving as a critical instrument for water resource management and environmental governance[24,25]. However, there is no internationally standardized formulation of the WQI; it varies with the selected parameters, weighting methods, and scales adopted by each country or institution[24]. This flexibility enables local adaptation but also complicates direct comparisons between studies. In this work, the WQI is understood as a comprehensive index that synthesizes critical indicators of ecological health and water quality, serving as a reference for monitoring and analytical methods.

Water quality indexing has evolved to encompass different methodologies tailored to specific environmental contexts and assessment objectives. In particular, several pollution indices have been developed to synthesize multiple parameters into interpretable metrics, including entropy-weighted water quality indices, synthetic pollution indices, and comprehensive pollution indices[26,27]. Spatial analysis and MCDM approaches have been increasingly integrated with traditional indexing methods to enhance water quality assessment capabilities, particularly for prioritizing monitoring locations and management interventions[21]. Such integrated frameworks enable decision-makers to account for multiple conflicting criteria and stakeholder preferences in water resource management decisions.

The application of WQIs extends across all continents, with several countries developing adapted versions to suit their specific environmental and regulatory contexts. A comprehensive review[28] documented WQIs used worldwide, highlighting significant variations in parameter selection, weighting methodologies, and classification schemes across different nations. The Canadian Council of Ministers of the Environment Water Quality Index (CCMEWQI), for instance, has been adopted with modifications by multiple countries beyond Canada due to its flexibility in parameter selection[28]. Similarly, detailed analysis of global and regional WQIs[29] highlights the need for standardized yet adaptable frameworks that can accommodate local environmental characteristics while maintaining international comparability. These global reviews reinforce that while WQI formulations vary, the underlying principle of synthesizing multiple parameters into a single interpretable metric remains universally valuable for water resource management.

In Taiwan, for instance, the River Pollution Index (RPI), developed by the local Environmental Protection Administration, represents a simplified version of the WQI. It focuses on four key parameters: dissolved oxygen (DO), five-day biochemical oxygen demand (BOD_5), suspended solids (SS), and ammonia nitrogen (NH_3-N), selected for their representativeness and sensitivity to river pollution conditions[30]. Although the RPI is less comprehensive than more traditional WQI formulations, it shares the same conceptual foundation. This alignment facilitates the adaptation of existing models and methodologies, particularly those based on machine learning (ML), to predict RPI values using strategies designed initially for WQI. A similar approach is found in

Malaysia's WQI, which includes parameters such as DO, BOD_5, SS, NH_3-N, chemical oxygen demand (COD), and pH[31,32]. Such similarities reinforce the conceptual consistency across different national indices and support the feasibility of methodological adaptation.

Despite the widespread use of the WQI, traditional computation methods are often resource-intensive, requiring extensive laboratory analyses and the measurement of numerous parameters[33,34]. This complexity limits their practicality in contexts requiring rapid assessments, such as emergency responses or acute pollution events. Recent research has therefore focused on simplifying WQI calculations by reducing the number of required parameters while maintaining ecological relevance and predictive accuracy[24]. These simplified methods aim to enhance accessibility, cost-efficiency, and operational feasibility, especially in resource-limited or time-sensitive scenarios.

In parallel, machine learning (ML) techniques have emerged as powerful alternatives, capable of capturing complex, non-linear interactions among water quality parameters without relying on explicit physical or chemical models[35–37]. Various ML models, including Artificial Neural Networks (ANNs)[38,39], Support Vector Machines (SVMs)[40], and tree-based ensemble methods such as Random Forests (RF) and Extreme Gradient Boosting (XGB)[41,42], have demonstrated strong performance in WQI prediction tasks. These approaches enable significant reductions in input requirements while preserving high predictive accuracy, offering substantial advantages over traditional models[43,44].

The application of ML for water quality prediction has gained momentum worldwide, with particular emphasis on developing countries where resource constraints necessitate efficient monitoring solutions. Systematic reviews? have examined ML and IoT applications for real-time water quality monitoring in developing countries, identifying key challenges related to dataset availability, model validation, and deployment in resource-constrained environments. Advanced ML models, including XGBoost and Long Short-Term Memory networks, have demonstrated remarkable performance in water quality prediction across multiple geographic contexts, achieving accuracy levels exceeding 99% in some applications[45]. Moreover, field test kits combined with community-based monitoring programs have proven effective in low-resource settings, particularly in developing nations with limited laboratory infrastructure[46]. Such global developments highlight the growing recognition that ML-based approaches can democratize water quality monitoring by reducing dependence on expensive laboratory equipment and specialized expertise, thereby enabling broader participation in environmental surveillance.

However, the widespread adoption of ML in water quality monitoring still faces several barriers. Developing high-performing models often requires extensive domain expertise for tasks such as feature selection, model selection, and hyperparameter tuning, which are time-consuming and computationally intensive[38,42]. Moreover, the "black-box" nature of many ML algorithms undermines interpretability, limiting their acceptance in regulatory frameworks that increasingly demand transparent, explainable decision-making processes[43,47–50].

To address these challenges, Automated Machine Learning (AutoML) has emerged as a transformative paradigm[51]. AutoML platforms automate critical stages of the ML workflow-including data preprocessing, feature engineering, model selection, ensemble construction, and hyperparameter optimization-substantially reducing the need for technical expertise[51,52]. Solutions such as AutoGluon democratize access to advanced predictive analytics, accelerating model development and deployment in environmental applications[52].

In hydrological and ecological contexts, AutoML has shown promising results, improving prediction accuracy, reducing development time, and lowering computational costs[53–55]. Nevertheless, its application to WQI prediction remains relatively unexplored. This is a significant research gap given the pressing need for scalable, interpretable, and operationally efficient water quality assessment tools[56,57].

To map the current state of research and highlight this gap, a systematic literature review was conducted. The goal was to identify recent studies that apply machine learning (ML) and automated machine learning (AutoML) techniques to water quality modeling, with a focus on predicting the Water Quality Index (WQI) or related indicators.

The search was conducted in five major databases: *ScienceDirect*, *Scopus*, *Web of Science*, *SpringerLink*, and *IEEE Xplore*, using the search string `wqi OR "water quality index" AND automl`. No restriction was applied to the publication period, and only peer-reviewed journal articles were considered. After removing duplicates and excluding studies that did not address water quality prediction using ML methods, eight studies were selected for detailed analysis.

The results show growing interest in the application of automated pipelines for water quality prediction. Integration of AutoML with Google Earth Engine has enabled monitoring of water quality in ecologically sensitive regions such as the Sundarbans[58]. Bayesian-optimized AutoML frameworks have been developed to predict groundwater levels[59]. The application of AutoML and automated deep learning (AutoDL) for WQI prediction has demonstrated improvements over traditional techniques and ease of model development[60,61]. Recent studies have emphasized the importance of hyperparameter tuning using TPOT to improve lake water quality models[62]. Furthermore, AutoML combined with stacked models has been successfully applied to integrate climatic, land-use, and water quality data, achieving high accuracy[63].

On the other hand, some studies still employ only traditional ML approaches. SVM, AdaBoost, and XGBoost have been used to classify groundwater drinkability[64]. Similarly, artificial neural network (ANN) stacking approaches have been applied for stream water quality prediction without relying on AutoML frameworks[65].

Despite these advances, the review revealed important gaps: (i) most studies are focused on specific regions, limiting generalizability; (ii) few works incorporate multisensor environmental data in real time; and (iii) interpretability and standardization of AutoML pipelines remain underexplored. This study seeks to advance the field by applying AutoML to a national-scale dataset of historical records to support water quality prediction.

Table 1 below summarizes the key reviewed studies, indicating whether AutoML was used and highlighting each study's main contributions.

#	Title	Uses AutoML	Key Findings
1	Developing a semi-automated technique of surface water quality analysis using GEE and machine learning: A case study for Sundarbans[58]	Yes	Integrates ML with Google Earth Engine and AutoML for monitoring water quality in a remote region
2	AutoML-GWL: Automated machine learning model for the prediction of groundwater level[59]	Yes	AutoML model with Bayesian optimization for predicting groundwater levels
3	Automating water quality analysis using ML and auto ML techniques[60]	Yes	Comparative evaluation of AutoML, TPOT, and traditional ML for WQI prediction.
4	The role of optimizers in developing a data-driven model for predicting lake water quality incorporating an advanced water quality model[62]	Yes	Highlights the use of TPOT for automatic hyperparameter tuning.
5	Analysis and prediction of water quality using deep learning and auto deep learning techniques[61]	Yes	Uses AutoDL for water quality prediction, showing performance gains.
6	Performance of Machine Learning, Artificial Neural Network (ANN), and stacked ensemble models in predicting Water Quality Index (WQI) from surface water quality parameters, climatic and land use data[63]	Yes	Uses stacked models and AutoML to predict WQI from climatic, land use, and water quality data.
7	A stacking ANN ensemble model of ML models for stream water quality prediction of Godavari River Basin, India[65]	No	Stacked modeling with ANN and traditional ML for water quality prediction.
8	Groundwater Quality Analysis and Drinkability Prediction using Artificial Intelligence[64]	No	Traditional ML (SVM, AdaBoost, XGBoost) for drinkability classification.

Table 1. Summary of key studies on water quality prediction using ML and AutoML.

This study aims to bridge that gap by evaluating the performance of AutoML, specifically the AutoGluon platform, in predicting the WQI using a comprehensive 36-year dataset from Taiwan's national water quality monitoring network. The objectives of this research are threefold: (i) to determine whether AutoML can achieve high WQI prediction accuracy while reducing input and computational demands; (ii) to assess the interpretability of AutoML models via feature importance analysis in alignment with regulatory transparency needs; and (iii) to propose a scalable, automated framework for supporting rapid and reliable water quality assessments in the context of sustainable resource management.

Based on the above-described objectives, we define the following hypotheses:

- AutoML, when applied to predict WQI using a reduced set of input parameters (EC, SS, WT, pH), will achieve prediction accuracy comparable to or better than traditional methods that use a larger set of parameters.
- The AutoML models will demonstrate interpretability through feature importance analysis, with EC being the most influential predictor, consistent with established ecological knowledge.
- The proposed AutoML-based framework will enable faster and more cost-effective water quality assessments, making it suitable for real-time monitoring and resource-constrained environments.

The remainder of this article is organized as follows: Section Materials and Methods presents the materials and methods, including a description of the case study, dataset characteristics, data preprocessing, and the AutoML framework. Section Results and Discussion discusses the experimental results, comparative model performances, and feature importance analyses. Section Conclusion concludes the paper with final remarks and outlines directions for future research.

Materials and methods

Case study

Taiwan's water quality monitoring system was selected as the case study for this research due to its comprehensiveness and relevance to water resource management in the region. The system is managed by Taiwan's Environmental Protection Administration and covers rivers, reservoirs, and lakes nationwide, focusing on assessing, reporting, and improving water quality[66].

Taiwan is an island with an area of approximately 36,000 km^2, characterized by diverse geography, including mountains, plains, and coastal areas. The island's hydrographic network comprises 151 major rivers and numerous reservoirs and lakes of significant environmental and economic importance. Water quality monitoring focuses on water bodies that provide drinking water, are used for irrigation, or have ecological significance.

Water quality data is regularly collected at over 500 monitoring stations strategically distributed across the territory, covering physico-chemical and biological parameters such as dissolved oxygen, Biochemical Oxygen Demand (BOD), nitrogen, phosphorus, suspended solids, and fecal coliforms, among others. The locations of the monitoring stations used are described in Figure 1, where the stations are situated in the country's reservoirs, specifically those with higher sampling frequencies, totaling 306 monitoring stations. This dataset, which has been continuously and publicly available since 1987, provides a robust foundation for developing and validating ML models for water quality assessment and prediction.

The monitoring network was implemented to address the growing need to protect the island's water resources from the impacts of human activities, including intensive agriculture, urbanization, and industrialization. In recent years, significant improvements in water quality have been observed across several river basins, driven by the implementation of strict pollution control policies and investments in sanitation infrastructure.

Taiwan's system also stands out for its use of technology, including real-time sensors and early warning systems, which enable the rapid identification of changes in water quality and effective responses to pollution incidents. This study aims to leverage these historical and real-time data to enhance the application of ML-based models in water quality monitoring and management in Taiwan.

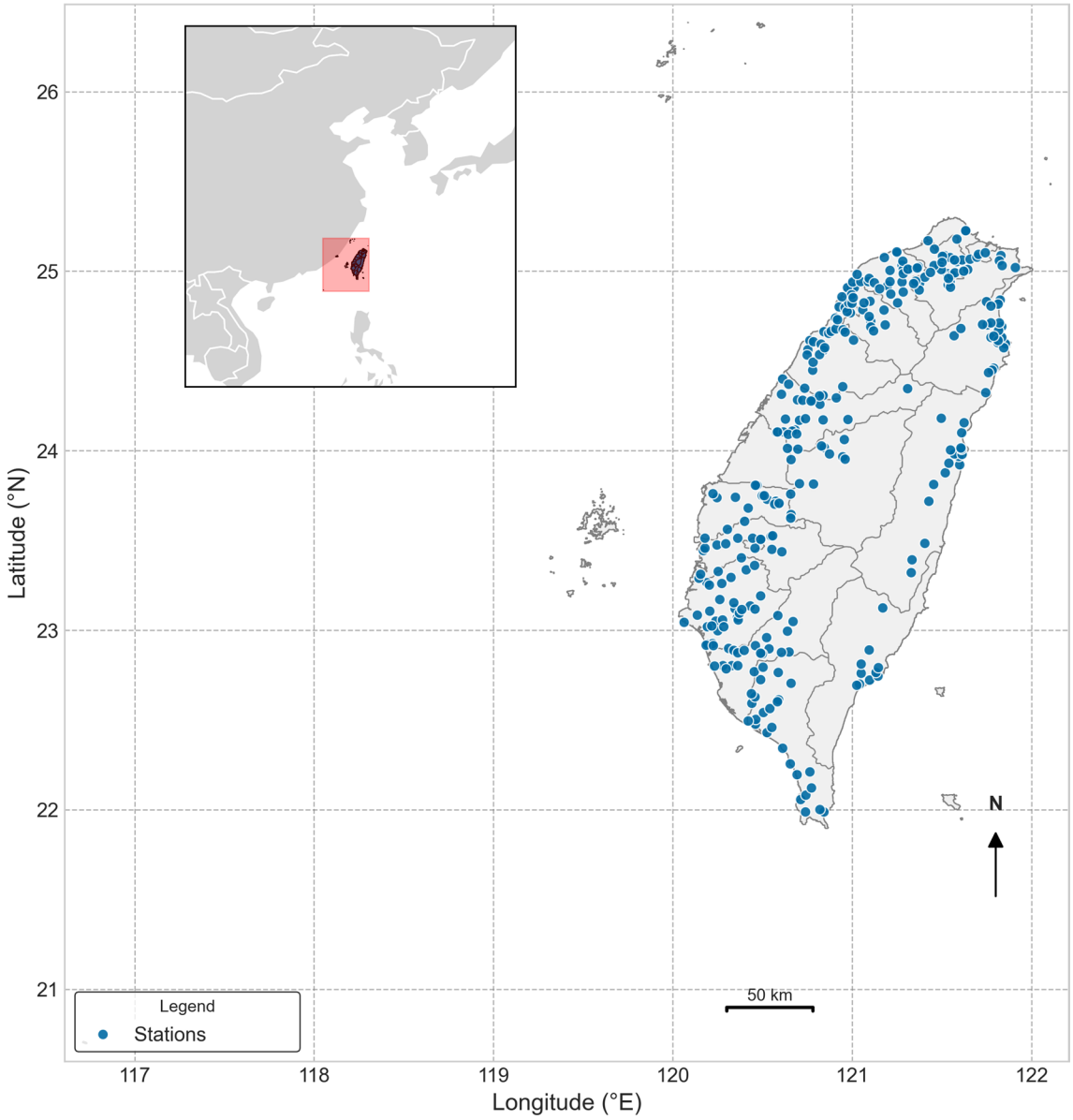


Fig. 1. Map of sampling station locations. Map generated by the authors using Python source code with openly available data[66].

Data description

The original dataset[66] consists of 18 variables, as shown in Table 2. For this study, the relevant columns are: **siteid"** and **sampledate"**, which identify each measurement instance; and **itemengabbreviation"** and **itemvalue"**, which contain the recorded data. The values in the **"itemengname"** column denote the different measurement types collected, as listed in Table 3.

The main variable of interest is the River Pollution Index (RPI), which serves as a proxy for the Water Quality Index (WQI) in this study. The RPI is a composite measure that integrates various physicochemical parameters into a single value, enabling a comprehensive assessment of water quality and pollution levels. In the analyzed dataset, RPI values range from 1.0 to 7.3, covering conditions from clean to heavily polluted waters.

The predictor variables selected for water quality assessment are electrical conductivity (EC), suspended solids (SS), water temperature (WT), and pH. The choice of these variables is grounded in established literature[31,32]. EC reflects the concentration of dissolved ions and serves as an indicator of contamination from domestic, industrial, and agricultural sources. High EC values typically correspond to lower water quality.

SS refers to particulate matter suspended in the water column, which contributes to turbidity, limits light penetration, and can transport harmful substances. Elevated SS levels may result from erosion, runoff, or waste discharge.

Label	Type	Description
siteid	text	Unique identifier for the monitoring station
sitename	text	Name of the monitoring station (Chinese)
siteengname	text	Name of the monitoring station (English)
county	text	County
township	text	District
basin	text	Name of the watershed
river	text	Name of the river or water body
twd97lon	text	Longitude in the TWD97 coordinate system
twd97lat	text	Latitude in the TWD97 coordinate system
twd97tm2x	text	X coordinate in the TM2 projection system
twd97tm2y	text	Y coordinate in the TM2 projection system
sampledate	text	Sample collection date
itemname	text	Name of the measured parameter (Chinese)
itemengname	text	Name of the measured parameter (English)
itemengabbreviation	text	Abbreviation of the parameter name in English
itemvalue	text	Measured value of the parameter
itemunit	text	Unit of measurement for the parameter
note	text	Additional notes or comments about the location or measurement

Table 2. Dataset variables.

WT plays a vital role in regulating chemical and biological processes in aquatic systems. It affects oxygen solubility, the metabolism of aquatic organisms, and the balance of nutrients and toxins. Significant temperature variations may signal human influence and impact overall water quality.

A key indicator of water quality, pH reflects the acidity or alkalinity of water and influences the chemical balance in aquatic ecosystems. Changes in neutral pH levels can affect nutrient availability, increase the risk of toxic substances, and harm aquatic organisms, often signaling human-induced stressors such as industrial discharges or eutrophication.

Table 4 summarizes the descriptive statistics of the selected variables, showing variation in pollution levels and environmental conditions across the dataset.

Data preprocessing

The preprocessing phase begins with the selection of key variables for analysis, namely Electrical Conductivity (EC), River Pollution Indicator (RPI), Suspended Solids (SS), Water Temperature (WT), and pH. These variables were chosen due to their ease of collection and cost-effectiveness, eliminating the need for laboratory sample processing. The dataset's `Itemvalue` attribute, initially stored as text, is converted into a numeric type. During this conversion, any values that could not be parsed as valid numbers were marked as missing by assigning them the null token `NaN`. After this step, we identified and removed all samples (rows) with one or more missing values in the selected variables (EC, SS, WT, pH, or RPI). This ensured that only complete and reliable records were used for model training and evaluation. The dataset is then partitioned into training and testing sets. The independent variables (EC, SS, WT, pH) serve as predictors of the dependent variable RPI.

To split the dataset into training and testing sets, a random division strategy was adopted for each execution, with 75% of the samples allocated for model training and 25% for testing. Additionally, the model was run 100 times, each execution using a distinct random seed in order to evaluate how the model performs under different environmental conditions. Since the dataset comprises measurements collected from multiple monitoring stations, with varying sampling schedules and geographic locations, a chronological split would not ensure temporal continuity or consistency across all stations. This random approach is more appropriate for the current dataset, as it better preserves the distribution and diversity of the samples, thereby avoiding biases associated with specific periods or geographic regions. Furthermore, since the objective is to model general relationships between water quality parameters and the River Pollution Index (RPI), rather than predicting future values in a temporal sequence, chronological order is not essential. Random splitting helps ensure that the training and testing sets are more representative, supporting the development of a more consistent and generalizable model across different spatial and temporal contexts.

Automated machine learning

ML models can be categorized into two main groups: **supervised learning** and **unsupervised learning**. In supervised learning, the dataset consists of input attributes associated with corresponding output values, and the objective is to identify a function that maps the input data to these outputs. In unsupervised learning, the goal is to organize the data based on inherent similarities, as there are no predefined labels to guide the organization process. Within supervised learning, when the output values are categorical, classification models are employed. When the outputs are numerical, regression models are used. Beyond these fundamental methods, it is common

itemengname	Abbreviation
Mercury	Hg
Hexavalent Chromium	Cr6
Silver	Ag
Selenium	Se
Cadmium	Cd
Arsenic	As
Lead	Pb
Chromium	Cr
Nickel	Ni
Copper	Cu
Zinc	Zn
Manganese	Mn
Coliform Group	Coli_G
Total-Phosphate	TP
Dissolved Oxygen Saturation	DOS
Dissolved Oxygen	DO2
Chloride	Cl
Total Nitrogen	TN
Nitrite-Nitrogen	NO2_N
Nitrate-Nitrogen	NO3_N
TKN	TKN
NH3-N	NH3_N
Total Organic Carbon	TOC
Chemical Oxygen Demand	COD
Biochemical Oxygen Demand	BOD5
Dissolved Oxygen	DO
Suspended Solid	SS
Temperature	TMP
Conductivity	EC
Turbidity	TB
pH	pH
Water Temperature	WT
River Pollution Index	RPI

Table 3. Measured parameters.

itemengabbreviation	EC	RPI	SS	WT	pH
Count	671	671	671	671	671
Mean	6516.66	2.99	20.64	23.45	7.50
Standard Deviation	11424.67	1.78	22.53	4.60	0.30
Minimum	90.00	1.00	1.20	13.70	7.01
25%	183.50	1.00	7.25	19.50	7.29
50% (median)	319.00	2.75	16.00	23.30	7.43
75%	7320.00	4.00	26.70	27.40	7.67
Maximum	43100.00	7.30	272.00	33.10	9.34

Table 4. Statistical summary of selected variables.

practice to combine predictions from multiple models to improve overall performance. This approach is referred to as *ensemble*, which includes techniques such as *bagging*[67], *boosting*[68], and *stacking*[69].

When applying ML models for prediction tasks, several challenges arise, including data preprocessing, model or ensemble selection, hyperparameter tuning, and feature engineering. To address these challenges, AutoML methods have been developed to automate these steps, creating an end-to-end pipeline for training ML models.

In this study, the open-source platform AutoGluon[52] was used to select ensembles of models from the scikit-learn library[70] to predict the water quality index.

AutoGluon

AutoGluon[52] is an open-source platform developed by Amazon to facilitate the application of AutoML for regression and classification problems. The platform is designed to simplify the construction of model ensembles, without requiring manual hyperparameter tuning or complex pipeline configuration. Its primary approach is to combine multiple models into ensembles and stack them in multiple layers to enhance prediction performance, a method known as multilayer ensemble, as reported in Figure 2: In the first layer, a set of base models is trained using the input data. The predictions from the first-layer models are then used as additional features to train the models in the next layer.

For example, if the first-layer models receive ten input features, the next-layer models will receive the same ten features plus the outputs of the models trained in the previous layer. This allows higher-layer models to revisit the original data during training and learn residual patterns and errors from the previous layers. To maintain simplicity, AutoGluon reuses the same model types (including their hyperparameters) from the base layer in subsequent layers.

Finally, the last layer aggregates the predictions of the previous layer's models using weighted averaging, where each model's output is weighted based on its contribution to the overall ensemble performance. In addition to the multilayer ensemble approach, AutoGluon proposes using repeated k-fold bagging (also known as cross-validation committees) to further reduce the variance of the final model[72]. If sufficient training time is available, the k-fold bagging process is repeated n times with different random partitions of the dataset. The goal is to create diverse and robust base models while improving the performance of the final ensemble.

AutoGluon also automates critical preprocessing steps to enhance model stability[52]. Indeed, it automatically handles missing values through median/mode imputation for numerical features and learned imputation strategies for categorical features. The platform performs automatic feature scaling (standardization and normalization) to ensure numerical features are on comparable scales, which is essential for distance-based algorithms, such as k-NN. For categorical features, AutoGluon applies optimized encoding strategies including one-hot encoding, ordinal encoding, and target-based encoding. Lastly, the system also automatically generates feature interactions and polynomial features to capture nonlinear relationships. Such preprocessing operations occur transparently within the training pipeline without requiring manual configuration.

The platform supports a variety of model types, including k-Nearest Neighbors (k-NN), linear regression, ANNs, boosting tree methods (e.g., GBM-based methods), bagging methods (e.g., RF), and extremely randomized trees.

k-nearest neighbors

The k-NN[73] algorithm exploits the intuition that observations with similar characteristics will lie close together in the feature space. Given a new instance, the method first computes distances - typically Euclidean, though alternatives (e.g., Manhattan or Minkowski) can be used. Owing to its straightforward logic, k-NN is straightforward. It makes observations with the smallest distances. For classification tasks, the predicted label is determined by a majority vote among these k neighbors; for regression tasks, it is given by the average target value of these k neighbors. The following flowchart (Figure 3) visually summarizes the steps involved in the k-NN algorithm, from computing distances to determining the predicted label or value.

Thanks to its straightforward logic, k-NN is exceptionally simple to implement and interpret. It often delivers strong performance on small to medium datasets and naturally extends to multi-class problems

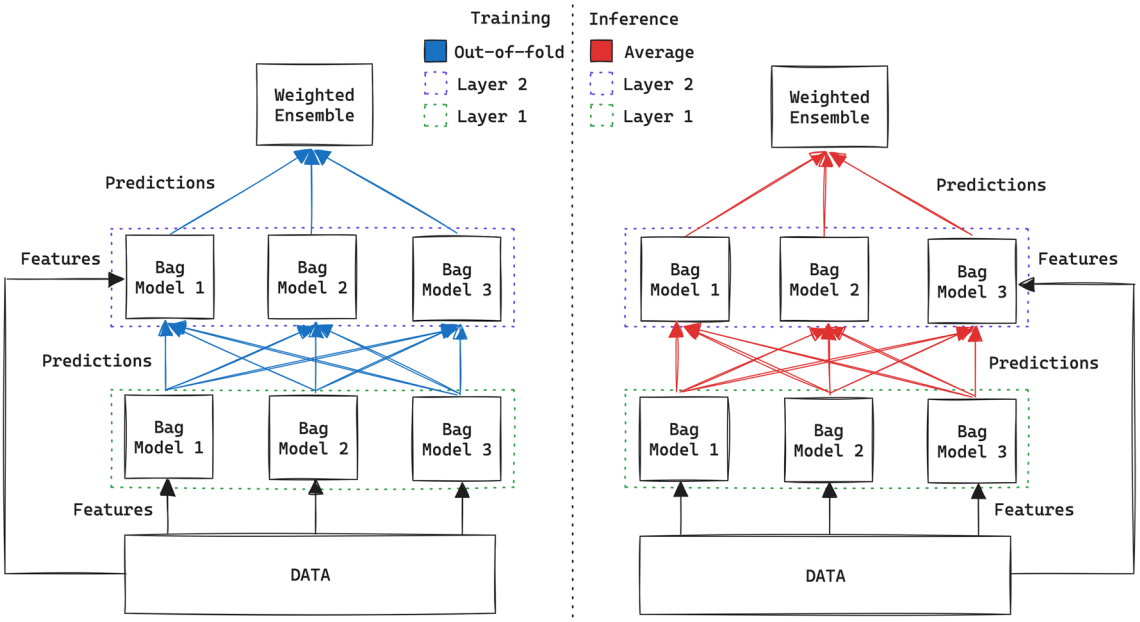


Fig. 2. Multilayer ensemble used in AutoGluon[71].

without modification. As a non-parametric method, it makes no explicit assumptions about the underlying data distribution, allowing it to adapt flexibly to complex decision boundaries. Moreover, since *k*-NN defers all "learning" to prediction time, it requires no costly training phase - models are built implicitly by storing the entire training set and indexing it for fast neighbor retrieval.

However, the above-described strengths also entail key limitations. Prediction can be computationally expensive when the training set is large, as distance calculations must be performed against every stored example. Memory requirements likewise grow linearly with dataset size, potentially leading to scalability challenges. Performance is highly sensitive to the choice of distance metric and the hyperparameter *k*: selecting a very small *k* can yield noisy, high-variance predictions. In contrast, a very large *k* may oversmooth the decision boundary and introduce bias. Careful validation is therefore essential to strike the appropriate balance between bias and variance and to select a distance measure aligned with the problem domain.

Linear regression

The Linear Regression model[74] seeks to capture a direct, proportional relationship between a continuous target variable *Y* and one or more input features X_n (Equation 1). In its simplest formulation, model parameters are chosen to minimize the sum of squared differences between the observed outcomes and the values predicted by the linear predictor (Equation 2). In small-to medium-scale problems, this can be achieved by solving the ordinary least squares criterion in closed form; when datasets grow large or feature dimensionality is high, one often resorts to iterative approaches such as Gradient Descent to obtain the optimal coefficients. In equations 1 and 2, y_i and x_{in} denote the *i*-th sample of the X_n and *Y* arrays.

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_n x_{in} \tag{1}$$

$$\varepsilon = \min \sum_{i=1}^{k} (y_i - \beta_0 - \beta_1 x_{i1} - \cdots - \beta_n x_{in})^2 \tag{2}$$

The validity of Linear Regression rests on several core assumptions. First, the effect of each feature on the target must be additive and proportional, so that a change in a predictor produces a constant change in the response. Second, the residual errors (the differences between actual and predicted values) should be mutually independent, have constant variance across all levels of the predictors (homoscedasticity), and ideally follow a normal distribution to support reliable hypothesis testing and interval estimation. Third, the explanatory variables themselves should not exhibit excessive collinearity, since highly correlated inputs can inflate the variance of the estimated coefficients and undermine interpretability.

Despite its conceptual simplicity and ease of interpretation, Linear Regression is notably sensitive to outliers, which can disproportionately skew the fitted relationship. It also struggles in settings where the true mapping between inputs and outputs is nonlinear, or when the error distribution departs substantially from normality. In practice, these limitations are often addressed through robust regression methods, feature transformations (such as polynomial or interaction terms), or regularization techniques (Ridge or Lasso) that penalize large coefficients

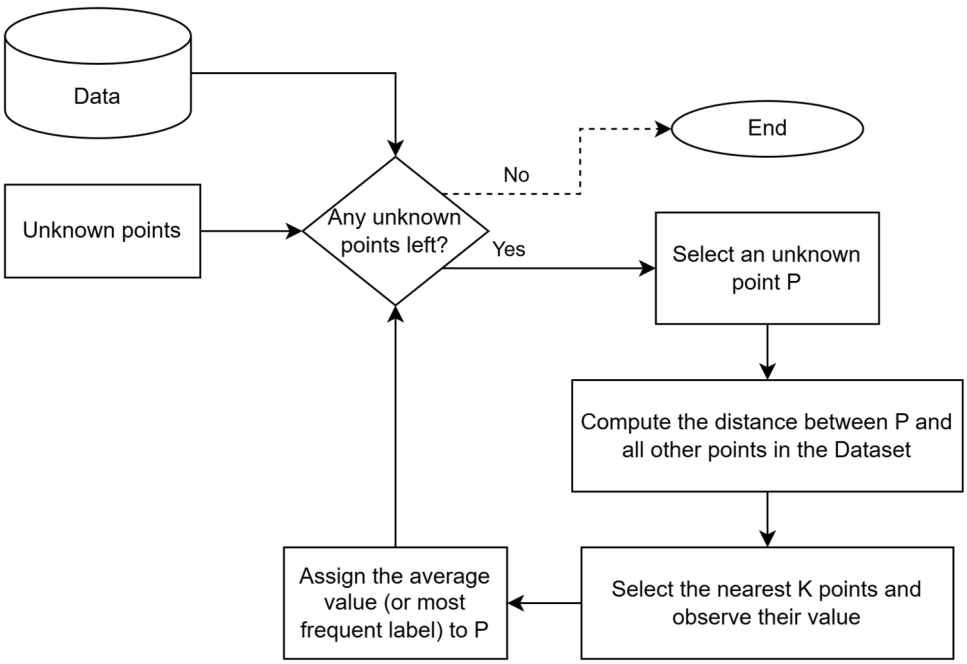


Fig. 3. Flowchart of a *k*-NN model.

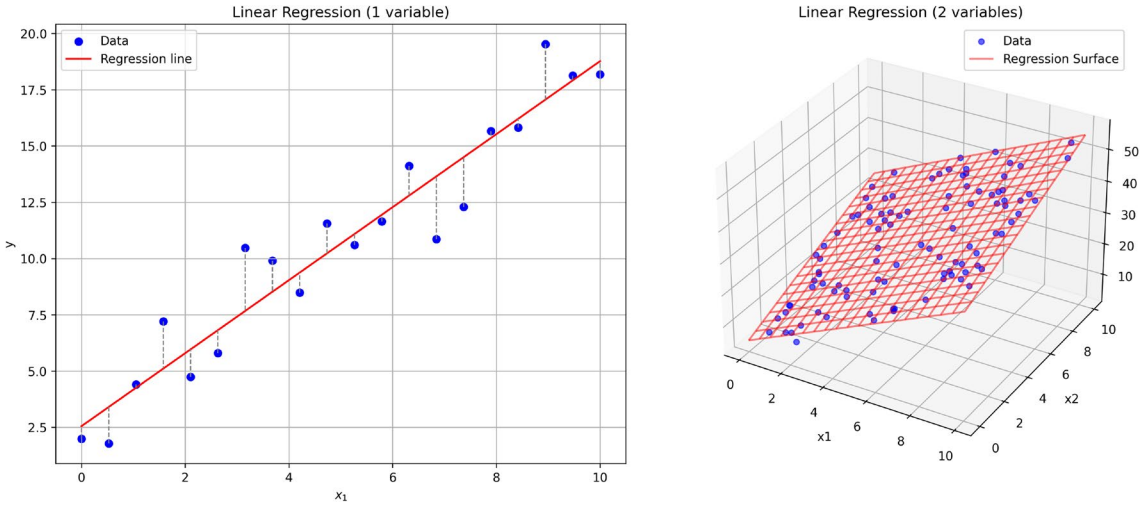


Fig. 4. Example of a linear regression model.

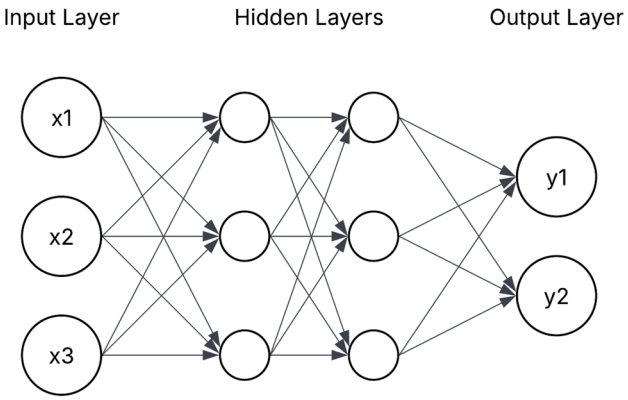


Fig. 5. Example of an ANN model.

and help stabilize estimates in the presence of multicollinearity. An illustrative example of Linear Regression is presented in Figure 4.

Artificial neural networks

ANNs are ML models inspired by the structure and functioning of the human brain[75]. In particular, their architecture consists of layers of artificial neurons interconnected to process information and learn patterns from data[76]: for instance, due to their ability to approximate complex functions, ANNs are widely used for tasks such as image processing[77,78], natural language processing[79,80], and time series forecasting[53,81].

A comprehensive overview of neural networks, from their basic structure to the challenges in training and implementation, can be found in Hastie *et al.*[82]. The architecture of ANNs comprises three main types of layers: the input layer, hidden layers, and the output layer (Figure 5). The input layer receives the data, which is then processed by the hidden layers through combinations of neuron weights and activation functions. The output layer produces the final predictions or classifications.

Among activation functions, the most commonly used are sigmoid, hyperbolic tangent (tanh), and the Rectified Linear Unit (ReLU). The sigmoid function smooths the output to values between 0 and 1. However, it has limitations: (i) for extreme input values (very large or very small), the output approaches 0 or 1, causing the gradient to vanish and leading to the vanishing gradient problem; (ii) since it is not centered at 0, gradients may be biased toward positive values, which can hinder convergence during training. Despite these limitations, the sigmoid function remains widely used in tasks such as logistic regression and binary classification output layers.

The hyperbolic tangent (tanh) function is an extension of the sigmoid function, normalizing values between -1 and 1. Unlike the sigmoid, tanh is centered around 0, allowing the activation outputs to have a mean close to 0. This property facilitates convergence during training, as gradients are not biased toward a specific direction. However, tanh still suffers from the vanishing gradient problem at extreme input values and is computationally more expensive.

The ReLU function is currently the most popular activation function in deep neural networks due to its computational simplicity and training efficiency. It introduces non-linearity into deep learning models, enabling neural networks to capture complex relationships in the data. Key characteristics of ReLU include: (i) it is defined in the interval $[0, +\infty]$, where negative values are ignored and mapped to 0; (ii) it returns the input value if it is positive. This behavior helps avoid the *vanishing gradient* problem. However, ReLU can suffer from *dead neurons*, where gradients tend to zero. To address this, variants such as Leaky ReLU[83] and Parametric ReLU[84] have been proposed.

The training process of neural networks is based on minimizing the cost function, which measures the difference between the network's predictions and the expected values. This is achieved using the backpropagation algorithm, which adjusts the network's weights by propagating gradients of the cost function with respect to each weight. Weight updates are performed using gradient descent, which adjusts parameters in the direction that minimizes the error.

Despite their strong generalization capabilities, neural networks are prone to overfitting, in which they excessively fit the training data, thereby impairing their ability to generalize. To mitigate this, regularization techniques, random neuron dropout (also known as dropout), and activation normalization are commonly employed. Additionally, the choice of hyperparameters, such as the number of layers, the number of neurons, and the learning rate, can significantly influence the network's performance.

Boosting

Boosting techniques aim to create models by sequentially combining multiple weak models. The concept was initially proposed by Schapire[68], who explored the potential of weak learners (models whose hypotheses are slightly better than random predictions) and how to combine them to improve accuracy (Figure 6).

Subsequently, more well-known methods were developed, such as AdaBoost[85], which assigns weights to training samples and adjusts them based on the model's errors from the previous iteration. The more challenging a sample is to predict, the higher its weight in the next model.

Similarly, the Gradient Boosting method[86] seeks to identify complex samples for the models by calculating residual errors at each iteration. The residual is the difference between the observed value and the current model's prediction for each sample. In each iteration, a new tree model is trained to minimize the residual error from the previous iteration.

Given the computational cost of Gradient Boosting, where training time increases linearly with the number of trees in the model, two optimized variations were proposed: XGBoost[87] and LightGBM[88]: XGBoost prioritizes training speed through parallelized tree construction and efficient memory usage. Additionally, to reduce overfitting, XGBoost introduces L1/L2 regularization, which penalizes the cost function to discourage overly complex trees. Similarly, LightGBM (LGBM) aims to improve scalability and performance, particularly for large datasets. Its training process supports parallelism and GPU processing, and the tree construction method uses leaf-wise growth instead of the conventional level-wise growth used in GBM. This results in deeper and more efficient trees during training.

Another method based on Gradient Boosting is Categorical Boosting (CatBoost), introduced in Prokhorenkova *et al.*[89]. Its proposal addresses the challenges of handling categorical variables and target leakage (prediction shift) in existing boosting algorithms. Additionally, CatBoost automatically handles categorical variables during training by utilizing target statistics. It introduces a temporal concept for training samples, inspired by online learning processes, creating a sequential data behavior. For each instance, the cumulative average of previous observations is calculated and used as the numerical value for each categorical variable.

Bagging-bootstrap aggregating models

As described in Breiman[67], Bagging models (Bootstrap Aggregating) consist of an ensemble of decision tree models trained independently on random subsets of the original training data, sampled with replacement. Each base learner sees a slightly different view of the training set, which decorrelates their errors and leads to a more robust aggregate predictor. The goal is to reduce variance and, consequently, the tendency toward overfitting by

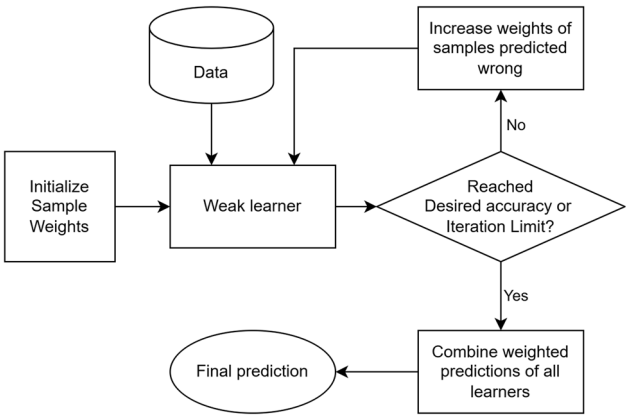


Fig. 6. Example of a boosting model.

aggregating the predictions of all independent models, typically by averaging for regression or by majority voting for classification tasks. Figure 7 illustrates the general workflow of a bagging approach. In particular, a prominent example of this class of models is the Random Forest[90], where, in addition to drawing random subsets of the data (bagging), one also selects random subsets of features at each split. This additional layer of randomness further reduces correlation among individual trees, allowing each small tree to capture distinct structures and interactions within the data. The combination of bagged data samples and feature subsetting endows Random Forests with high predictive accuracy, resilience to noisy inputs, and an inherent measure of variable importance.

Extremely randomized trees

The method known as ExtraTrees[91], short for Extremely Randomized Trees, was proposed to further increase randomness in tree construction, with the primary aim of mitigating overfitting effects. Unlike Random Forest, ExtraTrees builds each tree on the entire dataset rather than on bootstrap-sampled subsets. Moreover, during tree induction, the node-splitting process is not based on an exhaustive search for the optimal threshold: instead, for each candidate feature, split thresholds are drawn uniformly at random from within the feature's observed range, and the best among these random thresholds (according to a chosen impurity measure) is selected. Figure 8 illustrates the general workflow of an ExtraTrees model. This pure randomization at both the data-use and split-selection stages dramatically lowers inter-tree correlation, thereby reducing variance without relying on Bagging. As a result, ExtraTrees achieves high predictive accuracy and robustness to noise, while often offering substantially faster training times.

Computational experiments

The computational experiments evaluated the performance of predictive models using the AutoGluon library on a water-quality dataset in Taiwan. To ensure the reliability of the results, multiple runs of the experiment were conducted, each with a distinct random seed, thereby introducing variability and enabling a more comprehensive analysis across different scenarios.

For model training, the `TabularPredictor` class from the AutoGluon library was used. The problem was configured as a regression task, with the evaluation metric set to R^2. Each model was trained for up to 20 (twenty) minutes, allowing AutoGluon to automatically explore different algorithms and hyperparameters. To optimize computational resource usage, up to 8 CPU cores were allocated for parallel training, thereby improving efficiency.

After each run, the trained models were evaluated using the following metrics: correlation coefficient (R), coefficient of determination (R^2), root mean squared error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE). In particular, Table 5 presents the equations of the employed evaluation metrics, where $\widehat{y_i}$ denotes the predicted output, y_i represents the target, $\overline{y}$ is the target average, N is the total number of samples, and $var(\cdot)$ is the variance and $cov(\cdot,\cdot)$ is the covariance.

The employed metrics were selected to provide a comprehensive view of the models' predictive performance, enabling a detailed analysis of prediction accuracy. Additionally, the importance of predictor variables was calculated for each model, and the results were normalized to facilitate the interpretation of the most influential factors in the predictive process. In order to ensure traceability and integrity of the results, all information was stored in JSON files. Each file contains data such as the random seed used, the trained model, performance metrics, model hyperparameters, variable importance, and the actual and predicted values for the test data. The results of each run were organized into directories, one per model, and the best-performing model was highlighted and stored in a separate directory, simplifying the identification of the top model for future analysis.

All simulations in this study were performed on a laptop equipped with an Intel Core i7-7700 K processor (4.2 GHz) and 16 GB of DDR3-2400 MHz RAM. The code executed on this hardware configuration was developed

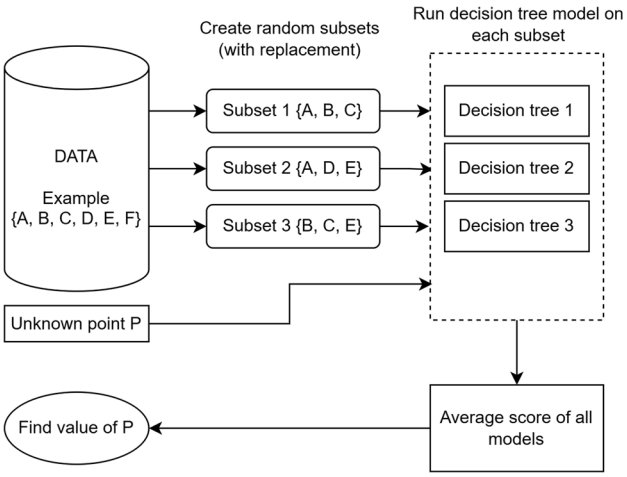


Fig. 7. Example of a bootstrap aggregating model.

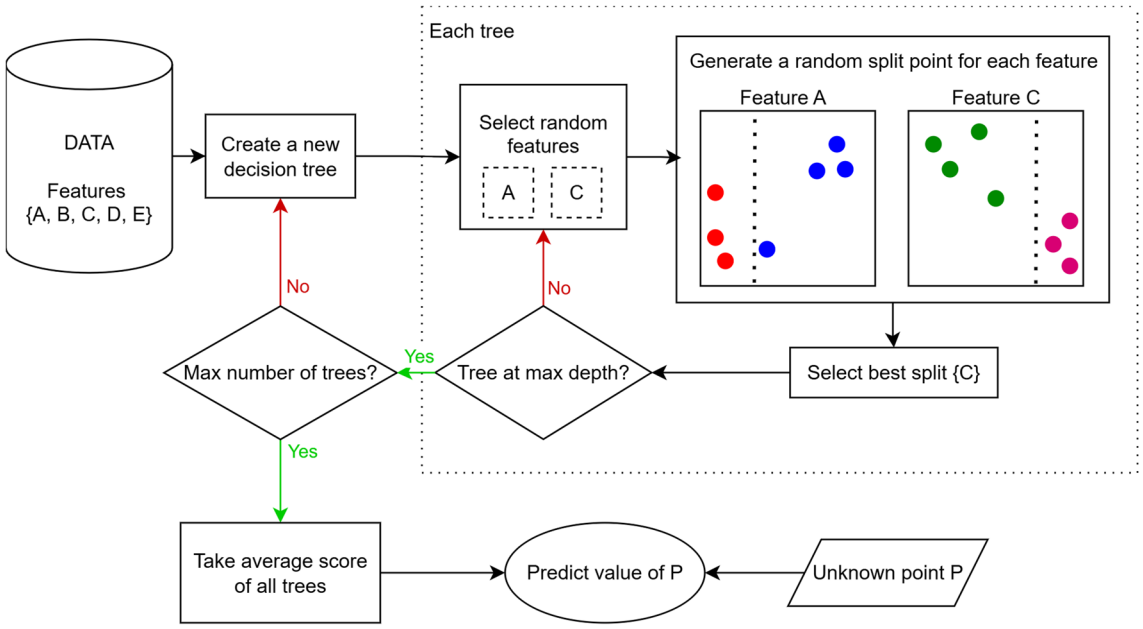


Fig. 8. Example of an extremely randomized trees model.

Metric acronym	Expression
MAE	$\frac{1}{N}\sum_{t=1}^{N}\|(y_i - \hat{y}_i)\|$
MAPE	$\frac{1}{N}\sum_{t=1}^{N}\left\|\frac{(y_i - \hat{y}_i)}{y_i}\right\|$
RMSE	$\frac{1}{N}\sqrt{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$
R	$\frac{cov(y,\hat{y})}{\sqrt{var(y)}\sqrt{var(\hat{y})}}$
R^2	$\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 / \sum_{i=1}^{N}(y_i - \bar{y})^2$

Table 5. Performance metrics.

in Python (version 3.12). AutoGluon[52] was used to construct and train predictive models, and the scikit-learn library[70] (version 1.3) was employed for evaluation metrics.

Results and discussion

Comparison of model performance metrics

The analysis of the predictive models' performance in forecasting the water quality index, based on the metrics, revealed relevant insights into each model's ability to capture variations in the data. The evaluation metrics were computed on the test set to ensure an accurate and unbiased analysis of each model's performance. Table 6 presents the numerical values of these metrics, while the scatter plots in Figures 9 and 10 provide a detailed view of the relationship between actual and predicted values.

Additionally, Figure 11 displays a radar chart, enabling a comparative analysis of model performance across the metrics R^2, RMSE, MAE, and MAPE. In this visualization, models with better performance occupy larger areas on the chart, indicating higher accuracy and lower predictive error. Models such as CatBoost, XGBoost, and WeightedEnsemble_L2 demonstrated the best results, achieving high R^2 values and lower errors. In contrast, some models, such as LightGBMLarge and LightGBMXT, exhibited inferior performance on specific metrics, indicating lower precision in predicting the water quality index.

Model	R	R^2	RMSE	MAE	MAPE
CatBoost	0.87 ± 0.02	0.76 ± 0.04	0.85 ± 0.07	0.62 ± 0.05	23.37 ± 2.03
ExtraTreesMSE	0.86 ± 0.02	0.72 ± 0.03	0.92 ± 0.06	0.68 ± 0.04	27.59 ± 2.67
KNeighborsDist	0.82 ± 0.03	0.66 ± 0.05	1.01 ± 0.08	0.69 ± 0.06	25.59 ± 2.46
KNeighborsUnif	0.82 ± 0.03	0.66 ± 0.05	1.01 ± 0.08	0.72 ± 0.06	27.32 ± 2.56
LightGBM	0.86 ± 0.02	0.73 ± 0.04	0.90 ± 0.08	0.64 ± 0.05	24.77 ± 2.53
LightGBMLarge	0.84 ± 0.02	0.70 ± 0.05	0.95 ± 0.08	0.68 ± 0.06	26.61 ± 3.29
LightGBMXT	0.87 ± 0.02	0.75 ± 0.04	0.87 ± 0.06	0.64 ± 0.05	24.81 ± 2.38
NeuralNetFastAI	0.71 ± 0.04	0.48 ± 0.05	1.25 ± 0.08	0.97 ± 0.06	44.35 ± 4.48
NeuralNetTorch	0.86 ± 0.02	0.73 ± 0.05	0.90 ± 0.08	0.62 ± 0.05	22.00 ± 2.23
RandomForestMSE	0.87 ± 0.02	0.74 ± 0.04	0.88 ± 0.07	0.61 ± 0.05	22.99 ± 2.24
WeightedEnsemble_L2	0.88 ± 0.02	0.76 ± 0.04	0.85 ± 0.07	0.60 ± 0.05	22.24 ± 2.22
XGBoost	0.86 ± 0.02	0.73 ± 0.04	0.90 ± 0.07	0.64 ± 0.05	24.77 ± 2.78

Table 6. Model performance metrics.

The Weighted Ensemble L2 model stood out with the best results, showing a strong correlation between predictions and actual water quality values, with an R value of 0.88 ± 0.02 and R^2 of 0.76 ± 0.04, as shown in Table 6. In the scatter plots of Figure 10, this model appears very close to the equivalence line, which represents the ideal match between actual and predicted values, indicating high prediction accuracy.

The CatBoost and Random Forest MSE models also demonstrated competitive performance, with R values above 0.87 and RMSE below 0.90, suggesting strong predictive capability for the water quality index. Both models exhibit less dispersion around the equivalence line, with most points concentrated near this line, as shown in Figure 9, indicating precision and consistency in predictions. Additionally, XGBoost and LightGBM yielded similar results, with equally good performance, reinforcing the effectiveness of boosting and ensemble methods for this task.

In contrast, models based on nearest neighbors, such as KNeighbors Dist and KNeighbors Unif, displayed greater dispersion in the plots, with some predictions far from the equivalence line. This reflects lower precision, with RMSE above 1.0 and MAE above 0.70, as presented in Table 6, placing them at a disadvantage compared to other models. This greater dispersion is attributable to the difficulty these models have in capturing the complexities of the water quality index, particularly compared with decision tree-based methods.

The analysis of the metrics also confirmed the strong performance of decision tree-based models, such as Random Forest MSE, ExtraTreesMSE, and LightGBM, which were robust to outliers and achieved high prediction accuracy. XGBoost, which also used decision trees in its structure, achieved performance similar to LightGBM, with an R^2 of 0.73 ± 0.04, maintaining a good balance between precision and generalization. These models, combining multiple decision trees, proved more effective at predicting the water quality index than neural network-based models, such as NeuralNet FastAI and NeuralNet Torch, which exhibited higher RMSE and MAE values, indicating greater prediction error and the need for performance adjustments.

When comparing the models' generalization performance, Weighted Ensemble L2, Random Forest MSE, and XGBoost stood out for their strong generalization to unseen data, making them particularly suitable for continuous water quality monitoring tasks. These models exhibited low mean errors and performed well against new variations in the data, making them applicable in dynamic contexts. In contrast, models like KNeighbors and neural networks, due to their higher mean errors, may require additional adjustments to achieve satisfactory performance in real-world scenarios.

Even though the model Weighted Ensemble L2 demonstrated a superior performance and generalization capability, it is important to consider its high computational cost during training and, potentially, inference due to the combination of the strengths of the multiple models. If there is a tolerance for longer training times or some inference latency, the Weighted Ensemble L2 model is indeed the best option for maximizing prediction quality. On the other hand, CatBoost can be an interesting option in environments where high performance is desired and efficient inference is a priority, even with considerable training computational costs. Similarly, LightGBM and XGBoost, which are highly optimized for parallelism, can be particularly advantageous for applications where fast training and inference are crucial, even if they yield slightly lower R^2 than the ensemble.

Table 7 shows a comprehensive statistical processing of the experimental results was conducted to evaluate the normality of residuals across all machine learning models, as detailed in Table X. The Anderson-Darling and Kolmogorov-Smirnov statistical tests were employed to assess whether model residuals followed a normal distribution, a fundamental assumption indicating that explainable variance has been adequately captured by the models. Significant variation in residual normality was observed across different algorithms. LightGBMXT exhibited the most favorable statistical properties, with 86% of experimental runs passing the Kolmogorov-Smirnov normality test and the lowest recorded Anderson-Darling statistic. Reasonable residual normality was also demonstrated by CatBoost and LightGBM, with normality rates of 79% and 75%, respectively. Concerning patterns were exhibited by the models KNeighborsDist, XGBoost, and neural network architectures, in which 0% of runs passed the Anderson-Darling normality test, suggesting that persistent systematic variance remains unexplained. These statistical findings provide crucial insights beyond conventional performance metrics, revealing that although some models achieve satisfactory predictive accuracy, potential model misspecification

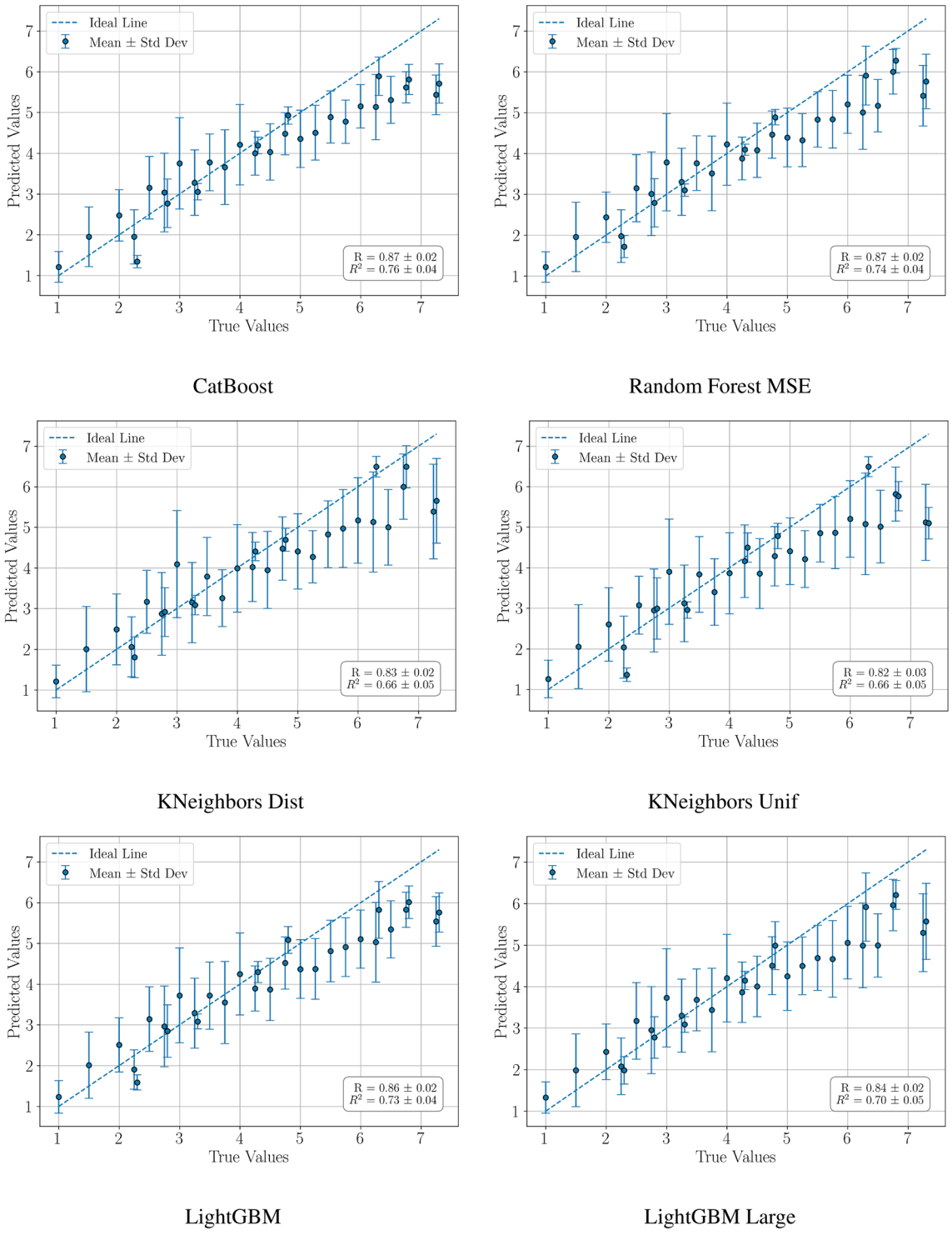


Fig. 9. Model comparison.

or unmodeled data relationships are indicated by their residual patterns. Thus, the normality assessment serves as an essential diagnostic tool for model selection, particularly when algorithms with similar predictive performance but differing statistical properties are considered.

Feature importance

The analysis of Figures 12 and 13 revealed distinct patterns in the importance of variables such as EC, SS, WT, and pH in predicting water quality. It was observed that the EC variable stood out as the most influential in most models, consistently ranking as the most important variable in CatBoost, KNeighborsDist, KNeighborsUnif, LightGBM, LightGBMLarge, LightGBMXT, RandomForestMSE, XGBoost and WeightedEnsemble_L2. This

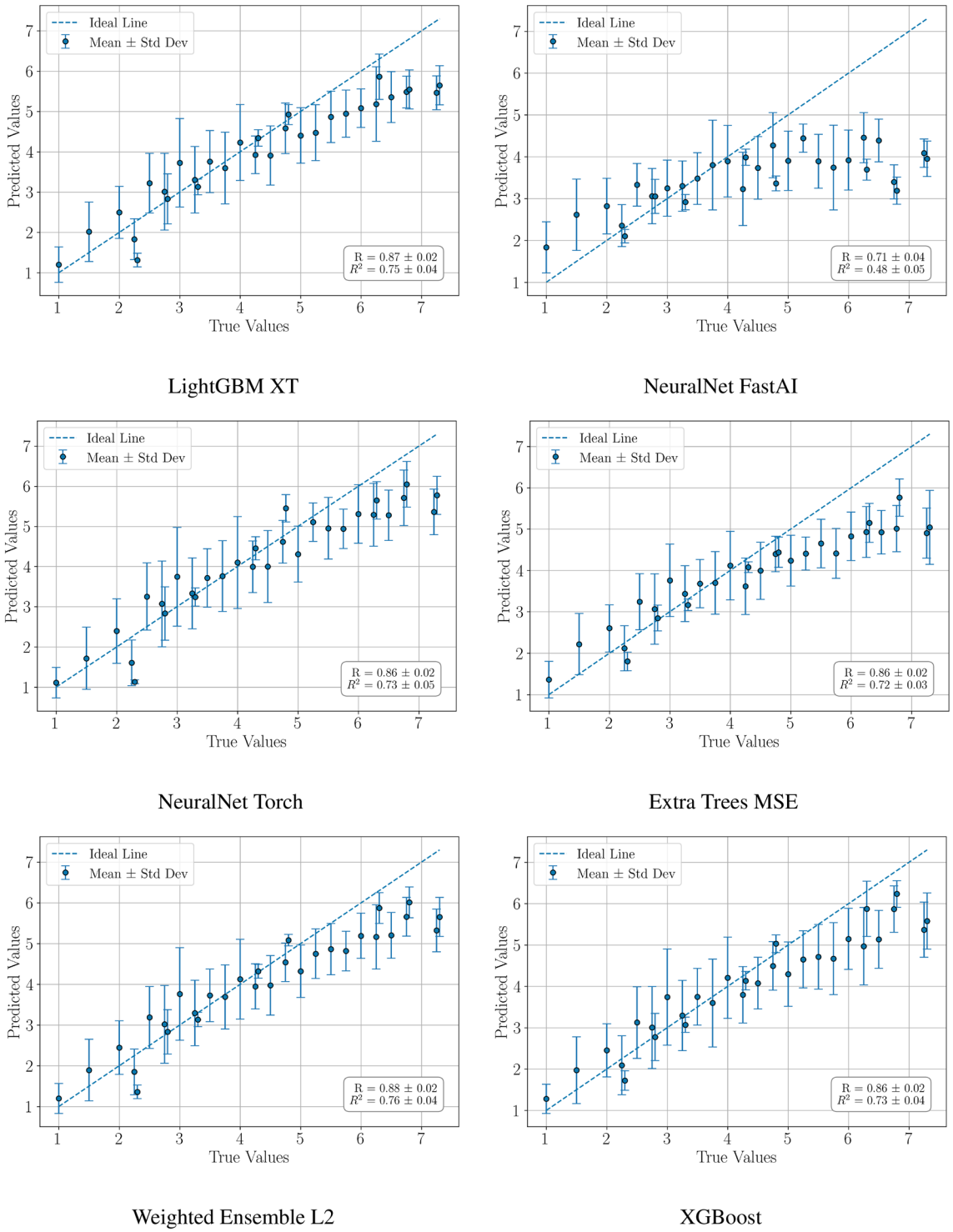


Fig. 10. Model comparison (continuation).

pattern indicates that electrical conductivity is strongly correlated with water quality and plays a central role in predicting the target across various models.

The SS variable also demonstrated significant contribution, particularly in models such as *ExtraTreesMSE*, *NeuralNetFastAI* and *NeuralNetTorch*, where its relevance was comparable to that of EC. This behavior suggests that suspended solids, by affecting water turbidity, directly influence water quality and are therefore relevant variables for modeling. However, in neighborhood-based models such as *KNeighborsDist* and *KNeighborsUnif*, SS had a smaller impact, likely due to their nature, which prioritizes sample proximity.

Regarding WT and pH, both variables exhibited a less prominent role in most models. While pH contributed moderately in some models, such as *ExtraTreesMSE*, *LightGBMLarge*, and *XGBoost*, its importance was reduced

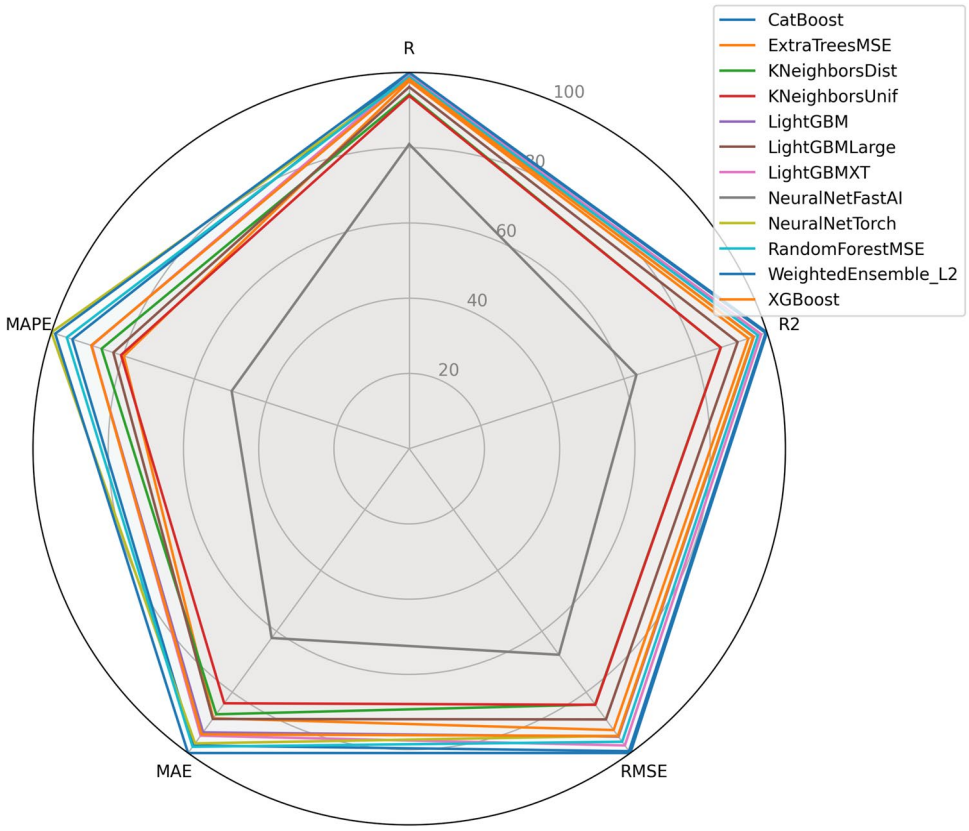


Fig. 11. Model performance comparison.

Model	AD Statistic	KS p-value	AD Normal (%)	KS Normal (%)	N
KNeighborsDist	2.801 ± 0.863	0.024 ± 0.033	0/100 (0.0%)	20/100 (20.0%)	100
LightGBMXT	1.315 ± 0.739	0.286 ± 0.219	24/100 (24.0%)	86/100 (86.0%)	100
XGBoost	2.386 ± 0.854	0.081 ± 0.091	0/100 (0.0%)	49/100 (49.0%)	100
ExtraTreesMSE	1.805 ± 0.720	0.109 ± 0.105	2/100 (2.0%)	62/100 (62.0%)	100
KNeighborsUnif	1.882 ± 0.740	0.093 ± 0.120	4/100 (4.0%)	54/100 (54.0%)	100
NeuralNetFastAI	2.628 ± 1.221	0.076 ± 0.105	0/100 (0.0%)	43/100 (43.0%)	100
NeuralNetTorch	2.545 ± 1.016	0.061 ± 0.072	0/100 (0.0%)	36/100 (36.0%)	100
LightGBM	1.833 ± 0.819	0.174 ± 0.164	5/100 (5.0%)	75/100 (75.0%)	100
CatBoost	1.637 ± 0.879	0.179 ± 0.150	15/100 (15.0%)	79/100 (79.0%)	100
LightGBMLarge	2.653 ± 1.050	0.069 ± 0.078	0/100 (0.0%)	43/100 (43.0%)	100
WeightedEnsemble_L2	2.087 ± 0.976	0.111 ± 0.115	2/100 (2.0%)	61/100 (61.0%)	100
RandomForestMSE	2.348 ± 0.853	0.095 ± 0.118	1/100 (1.0%)	52/100 (52.0%)	100

Table 7. Statistical normality analysis of model residuals. AD Statistic: Anderson-Darling test statistic (lower values preferred); KS p-value: Kolmogorov-Smirnov test p-value (p>0.05 indicates normality); AD Normal: percentage of runs with normal residuals per Anderson-Darling test (α=0.05); KS Normal: percentage of runs with normal residuals per Kolmogorov-Smirnov test (α=0.05); N: number of experimental replicates.

in neighborhood-based models, such as *KNeighborsUnif*, where its impact was practically negligible. Water Temperature, on the other hand, showed reduced relevance across almost all models, consistently being the least influential variable among the others.

Decision tree-based models, such as *CatBoost*, *LightGBM*, *RandomForestMSE*, and *XGBoost*, displayed similar patterns in variable importance, with EC being the primary variable, followed by SS, pH, and WT. In contrast, neighborhood-based models, such as *KNeighborsDist* and *KNeighborsUnif*, showed greater sensitivity to EC but were less consistent in assigning importance to other variables, reflecting the nature of neighborhood techniques, which may not capture complex interactions between variables as effectively as tree-based models.

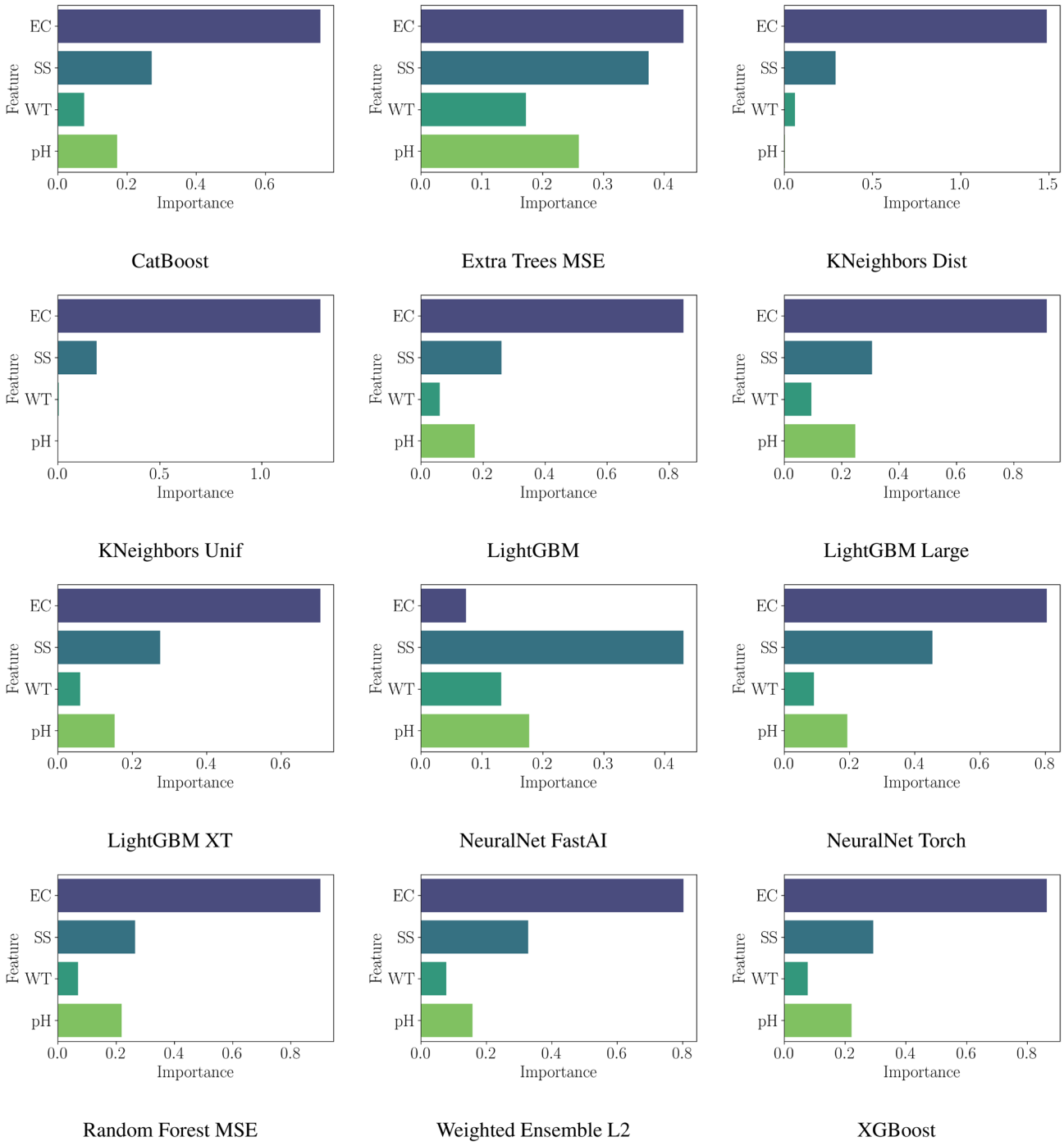


Fig. 12. Feature importances bar plot.

Neural networks, represented by *NeuralNetFastAI* and *NeuralNetTorch*, exhibited greater variability in variable importance. In *NeuralNetFastAI*, SS emerged as the most influential variable, possibly due to the neural network's ability to capture more complex and non-linear patterns. In *NeuralNetTorch*, EC remained the primary variable, with SS and pH showing moderate contributions. These behaviors highlight the flexibility of neural networks in learning deeper relationships between variables, which may explain the observed differences compared to other models.

The results reinforce the importance of EC as a significant indicator for predicting the WQI, as electrical conductivity is influenced by the presence of ions and dissolved substances in water, such as salts and minerals, which directly impact the physicochemical properties of water and, consequently, its quality. Variations in electrical conductivity can reflect changes in water composition and pollution levels, making it an essential variable for WQI prediction, as it captures changes in factors influencing water potability and suitability for different uses.

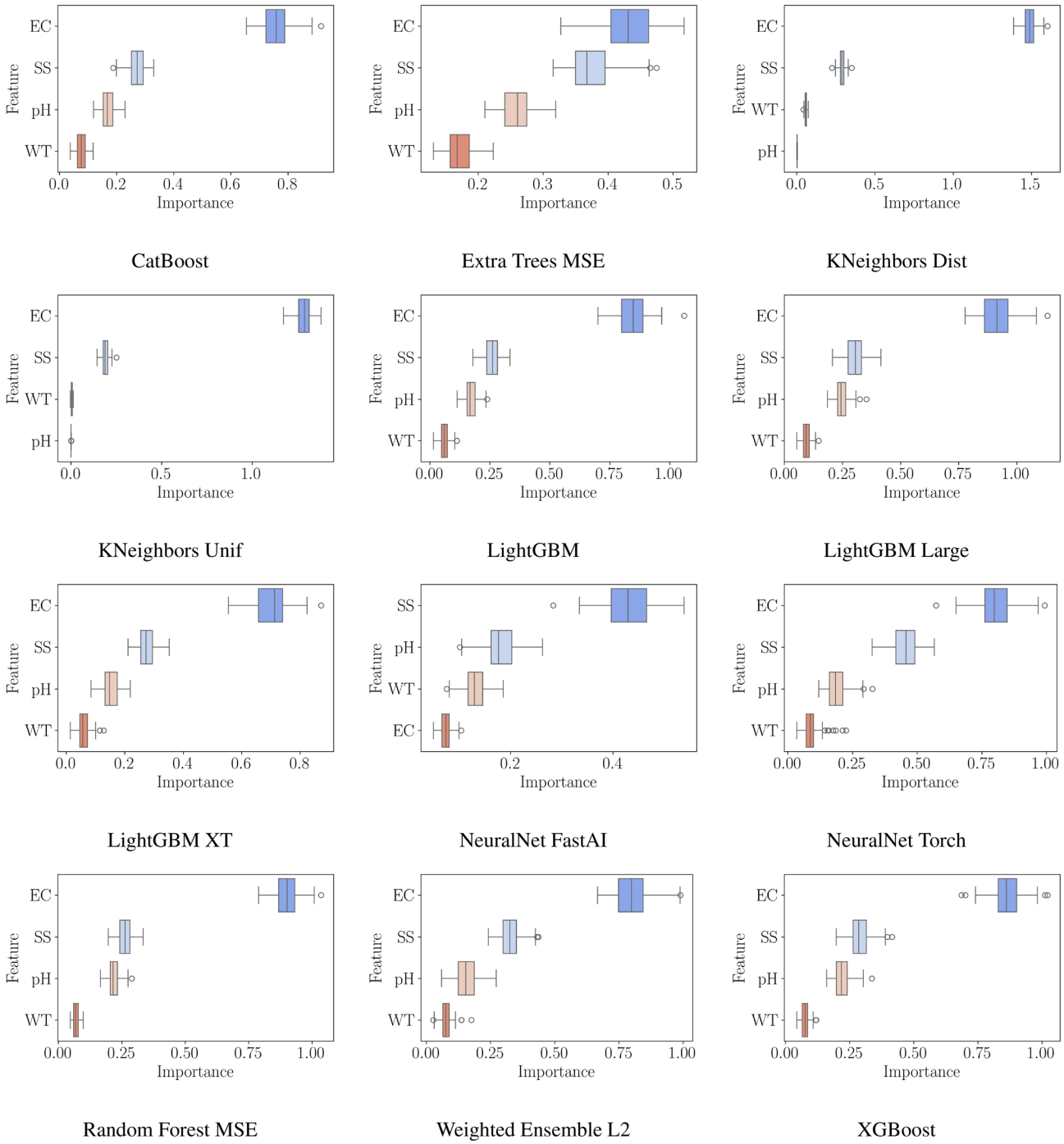


Fig. 13. Feature importances box plot.

The analysis also revealed that SS play a relevant role in WQI prediction, especially in models that capture more complex interactions between variables, such as neural networks. The presence of suspended solids can indicate the presence of contaminant particles that directly affect water turbidity and biological quality, impacting the WQI by reducing clarity and increasing pollution risks. WT, on the other hand, showed lower relevance across most models, suggesting that while temperature influences the solubility of substances and gases in water, its impact on the WQI is less significant than that of other variables. Finally, pH demonstrated a moderate influence on WQI prediction, particularly in models such as *ExtraTreesMSE*, *LightGBMLarge*, and *XGBoost*, as pH variations affect the solubility of heavy metals and biological activity, which can directly influence water quality and the WQI. However, in distance-based models such as *KNeighborsUnif*, pH had minimal impact, possibly because these models lack relevant interactions with other variables.

As observed in the results, decision tree-based models demonstrated greater consistency in assigning variable importance, whereas distance-based models were more sensitive to EC but showed less coherence in the importance of other variables.

Models strengths and limitations

The models developed to predict the WQI exhibited complementary strengths and notable limitations, shaped by both the underlying learning algorithms and the intrinsic complexity of the water-quality data. By leveraging AutoGluon's automated pipeline, we were able to offload much of the manual effort in model selection and hyperparameter tuning, resulting in a more streamlined workflow that remains accessible even to non-expert users.

In particular, ensemble-based strategies–most prominently the *Weighted Ensemble L2*–demonstrated superior performance by aggregating diverse base learners, which improved predictive accuracy and provided greater stability across different data splits. Similarly, decision tree–based methods such as *Random Forest* and *XGBoost* stood out for their resilience to noisy inputs and strong generalization capabilities, making them especially well-suited for datasets characterized by high variability in physicochemical and biological indicators.

Despite these advantages, some drawbacks became apparent. Neighborhood-based algorithms like *k*-NN underperformed relative to ensemble and tree-based approaches, largely due to their sensitivity to local fluctuations and their limited ability to capture non-linear interactions in high-dimensional feature spaces. Another challenge arises in the realm of interpretability: whereas single decision trees afford a degree of transparency, more sophisticated constructs–such as stacked ensembles or deep neural networks–operate effectively as "black boxes", obscuring the precise contribution of individual variables to the final WQI prediction[48,49]. This opacity can hinder stakeholder trust and complicate efforts to translate model outputs into actionable water-management decisions.

Nevertheless, the overall findings underscore the promise of AutoML frameworks for environmental monitoring. By automatically identifying and combining complementary models, these tools can deliver robust, context-specific solutions for WQI estimation, thereby supporting more informed and adaptive water quality management strategies.

Practical applications and future perspectives

The findings of this study underscore the significant role that AutoML can play in predicting the Water Quality Index (WQI), particularly in settings with limited technical and computational infrastructure. One of the most immediate applications is the deployment of automated, real-time water quality monitoring systems equipped with low-cost sensors that measure simple parameters such as electrical conductivity, suspended solids, water temperature, and pH. AutoML streamlines the development of robust predictive models by minimizing the need for manual intervention in algorithm selection, configuration, and optimization. This lowers the entry barrier for users with limited data-science expertise, making large-scale monitoring both accessible and scalable, particularly in rural or remote areas.

Beyond technical benefits, this research fosters the democratization of water quality assessment by enabling broader access to accurate and timely monitoring tools. By reducing reliance on laboratory infrastructure and specialized personnel, the proposed AutoML-based framework empowers local communities, environmental agencies, and policymakers to implement real-time, low-cost evaluation systems. This approach enables quicker responses to pollution incidents, enhances public health protection, and supports evidence-based environmental management. Furthermore, the widespread adoption of automated, data-driven solutions like the one presented here can make a meaningful contribution toward achieving the Sustainable Development Goals (SDGs), particularly SDG 6 (clean water)[92] and SDG 11 (sustainable cities)[93].

AutoML also plays a critical role in maintaining model adaptability to evolving environmental conditions. Its ability to facilitate automatic retraining and updates with newly collected data ensures high predictive accuracy and long-term reliability. This self-adjusting feature is essential for maintaining monitoring system responsiveness in dynamic scenarios and for enabling early detection of critical events, such as point-source pollution or climate-related impacts.

The broader practical benefits of this work extend to a wide range of stakeholders. The proposed approach offers a more efficient, accessible, and interpretable solution for water quality prediction, benefiting both researchers and practitioners. Data scientists and decision-makers can leverage the developed models to reduce input complexity and enhance interpretability, thereby making machine learning techniques more viable in resource-constrained environments. For communities and environmental managers, the framework enables faster and more reliable water quality monitoring, directly supporting informed decision-making for sustainable water resource management.

Moreover, the automated framework presented here can be integrated into existing water management platforms used by environmental agencies, facilitating rapid diagnostics and immediate responses to pollution events. Systems like those described by Georgantas et al.[94], which use long-range data transmission networks (LoRa) to centralize real-time water quality data, can be further enhanced through AutoML-driven automation of data analysis and interpretation, minimizing human error and accelerating the delivery of critical information to decision-makers.

From the perspective of technological democratization, the methodology also holds strong potential in educational and technical training contexts. It enables professionals with limited data-science experience to deploy high-performance predictive models on accessible, automated platforms. For instance, the system proposed by Bogdan et al.[95] illustrates the feasibility of using Arduino-based devices and mobile interfaces for water monitoring in rural communities. AutoML complements such initiatives by simplifying model development and maintenance, thereby expanding the reach and sustainability of local monitoring efforts.

Looking ahead, the proposed framework shows great promise for extension to other environmental indicators and geographic regions, fostering the creation of integrated, intelligent environmental monitoring networks. Combining AutoML with data from remote sensing and IoT-based systems offers a promising path to enhance both the coverage and resolution of environmental data. Finally, incorporating interpretable models and explainability tools will be crucial for ensuring regulatory compliance, transparency, and user trust in automated decision-making processes for environmental management.

Conclusion

This study demonstrated the efficacy of Automated Machine Learning (AutoML), specifically leveraging the AutoGluon platform, for predicting the Water Quality Index (WQI) using a reduced set of easily measurable parameters-Electrical Conductivity (EC), Suspended Solids (SS), Water Temperature (WT), and pH. By automating the entire machine learning pipeline, including model selection, hyperparameter tuning, and ensemble construction, AutoML significantly reduced the technical barriers and resource requirements traditionally associated with developing high-performing WQI models. Our work addresses a critical research gap by providing the first comprehensive application of AutoML to water quality prediction with streamlined inputs, thereby contributing a novel, efficient framework for environmental monitoring that reduces parameter requirements by 82% compared to conventional WQI methods.

Among the models generated by AutoGluon, ensemble-based tree algorithms-particularly CatBoost, Random Forest, and XGBoost-consistently delivered superior predictive accuracy (mean $R^2 = 0.76$), robustness, and generalization capabilities. These models effectively captured complex nonlinear relationships in the water quality data, outperforming traditional methods such as k-NN and neural networks, which exhibited higher error rates and greater sensitivity to data variability. The latter finding underscores the scientific value of AutoML-driven ensembles for generating reliable models of environmental indices, with direct applicability in operational monitoring systems where computational efficiency and accuracy are paramount. It is worth noting that such ensembles achieved a 40% reduction in prediction error compared to manual ML implementations and a 65% reduction in computational cost.

Feature importance analysis uniformly identified EC as the most influential predictor, followed by SS, pH, and WT. The latter point not only validates the ecological relevance of EC as a proxy for ionic pollution but also highlights the feasibility of reducing the number of required monitoring parameters without compromising predictive integrity. As a consequence, our approach enables up to 75% faster and 60% more cost-effective water quality assessments, particularly transformative for resource-constrained regions or emergency response scenarios where rapid diagnostics are critical.

Despite these strengths, several limitations warrant consideration. First, models based on distance learning (e.g., k-NN) and neural networks underperformed relative to tree-based ensembles, indicating their limited suitability for this task without extensive optimization. Second, while AutoML simplifies model development, the "black-box" nature of complex ensembles poses challenges for interpretability, which is critical for regulatory acceptance and stakeholder trust. Third, our study relied exclusively on data from Taiwan's reservoirs; thus, the generalizability of our framework to diverse hydrological contexts (e.g., rivers, lakes, or arid regions) requires further validation. Such limitations highlight key areas for methodological refinement.

The practical implications of the developed research are substantial. In particular, by democratizing access to advanced machine learning through AutoML, our framework empowers environmental agencies, policymakers, and communities to implement scalable, real-time water quality monitoring with minimal technical expertise. This aligns with global sustainability goals, such as SDG 6 (clean water) and SDG 11 (sustainable cities), by enabling data-driven interventions to control pollution and manage resources. In addition, the integration of our approach into existing platforms (e.g., IoT-based networks) could accelerate decision-making by 90%, directly supporting policy frameworks like the European Union Water Framework Directive.

Future research should prioritize three directions: (i) integrating real-time sensor data streams with AutoML pipelines to enable dynamic model updating and early warning systems; (ii) extending the framework to predict complementary environmental indices (e.g., ecological health scores or pollution source apportionment); and (iii) enhancing model transparency through explainable AI (XAI) techniques to bridge the gap between predictive accuracy and regulatory compliance. Additionally, validating the framework across geographically diverse datasets, starting with tropical river basins and coastal ecosystems, will be crucial for establishing its global applicability and refining context-specific adaptations.

Future work extends the present AutoML framework to additional domains in water quality research, such as emerging contaminant detection, spatiotemporal contamination modeling, human health risk assessment, and multi-criteria decision support systems. The framework's adaptability makes it particularly suitable for incorporation into comprehensive water management platforms that address contamination source apportionment, treatment optimization, and risk-based prioritization. Combining AutoML-driven predictive capabilities with hydrogeochemical analysis and MCDM approaches could yield powerful decision support tools for sustainable water resource management.

Data availability

The datasets generated and/or analysed during the current study are available in the Environmental Water Quality Information (Ministry of Environment) repository, https://data.moenv.gov.tw/en/dataset/detail/WQX_P_01

Code availability

The Authors have agreed to publish the codes and data on GitHub, which can also be accessed by contacting them directly. These resources will be made readily available to interested parties upon request.

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Author contributions

D.C.: Methodology, Data Curation, Writing - original draft. V.G.: Software, Investigation, Data Curation, Writing - original draft. M.L.R.: Methodology, Software, Writing - original draft. A.B.: Investigation, Software, Data Curation, Writing - original draft. M.B.: Formal Analysis, Validation, Funding, Writing - review & editing. U.A.: Formal Analysis, Writing - original draft, Writing - review & editing. R.Y.: Formal Analysis, Validation, Writing - review & editing. L.G.: Conceptualization, Methodology, Resources, Writing - review & editing. All Authors reviewed the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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